清華大學

收缩及徐变部分习题

3-2 (1) $\sigma_{c0}=\dfrac{\varepsilon_{sh}E_s}{1+\dfrac{\alpha_E\rho}{\nu}}=24.00\ MPa$　　$\sigma_0=\dfrac{\varepsilon_{sh}E_c}{\dfrac{1}{\alpha_E\rho}+\dfrac{1}{\nu}}=0.78\ MPa$

(2) 由于加N之间，混凝土已有0.78 MPa之拉应力，设加N后又产生 ε 的混凝土应变

则 $\sigma_c=\nu E_c(\varepsilon+\dfrac{\sigma_0}{E_c})$ ①

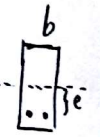
而钢筋因压应变，$\sigma_s=E_s\varepsilon-\sigma_{s0}$

在混凝土开裂前有 $\begin{cases}\sigma_c=f_t\\ \nu=0.5\end{cases}$ ⇒ 可由①式得出 $\varepsilon=127.6\mu$.

则 $\sigma_{s前}=E_s\varepsilon-\sigma_{s0}=1.34\ MPa$.　　$N_{cr}=\sigma_cA_c+\sigma_sA_s=81.27\ kN$

$\sigma_{sf}=\dfrac{N_{cr}}{A_s}=64.67\ MPa$.

3-3. 假设不对称配筋情况下，钢筋都布在中性轴同侧，且距中性轴距离为 e（设中性轴位置难）



设混凝土自由收缩应变 ε_{sh}

收缩时，$F_{c,s}=F_{t,c}\Rightarrow E_s\varepsilon_sA_s=A_c\sigma_c=\nu A_cE_c\varepsilon_0\Rightarrow\varepsilon_{c0}=\dfrac{\alpha_E\rho}{\nu}\cdot\varepsilon_s$ ①

由钢筋偏心受压，故其会对混凝土产生弯矩 $M=F_{c,s}\cdot e$

为使变形协调，要求出钢筋所在位置之混凝土弯曲应力 $\sigma_{c,M}=\dfrac{Me}{I_z}=\dfrac{12F_{c,s}e^2}{bh^3}$（对混凝土起拉伸作用）

故可求出 $\varepsilon_{c,M}=\dfrac{\sigma_{c,M}}{\nu E_c}$ ②

由变形协调　$\varepsilon_{c0}+\varepsilon_{c,M}+\varepsilon_s=\varepsilon_{sh}$　　代入①② 有 $\varepsilon_s=\dfrac{\varepsilon_{sh}}{1+\dfrac{\alpha_E\rho}{\nu}+\dfrac{12\alpha_E}{\nu}\dfrac{\rho}{1+\rho}(\dfrac{e}{h})^2}$

$\sigma_s=E_s\varepsilon_s=\dfrac{E_s\varepsilon_{sh}}{1+\dfrac{\alpha_E\rho}{\nu}+\dfrac{12\alpha_E}{\nu}+\dfrac{\rho}{1+\rho}(\dfrac{e}{h})^2}$

由变应力之线性分布　$\varepsilon_{c(y)}=\varepsilon_{c0}+\varepsilon_{c,M}\cdot\dfrac{y}{e}$　　$\sigma_{c(y)}=\nu E_c\varepsilon_c=\dfrac{E_s\rho\,(1+\dfrac{12ey}{(1+\rho)h^2})\varepsilon_{sh}}{1+\dfrac{\alpha_E\rho}{\nu}+\dfrac{12\alpha_E\rho}{\nu}\dfrac{\rho}{1+\rho}(\dfrac{e}{h})^2}$

由 扫描全能王 扫描创建

3-6.

(1) 由过镇海曲线 $\frac{\sigma_c}{f_c'} = 2.09\frac{\varepsilon}{1560} - 1.18\left(\frac{\varepsilon}{1560}\right)^2 + 0.09\left(\frac{\varepsilon}{1560}\right)^3$ ①

由轴力公式 $N = \sigma_s A_s + \sigma_c A_c$. ② 且 $\varepsilon_s = \varepsilon_c = \varepsilon$.

将①代入②有一元三次方程，代入 Matlab 求正根 $\varepsilon = 110.765\mu$

∴ 由① $\sigma_{c,0} = 3.56\,Mpa$ $\sigma_{s,0} = E_s\varepsilon = 22.15\,Mpa$

由 $\psi = 2$

$$\sigma_{c,t} = \sigma_{s,0}\left[1 + \frac{\psi}{1+\alpha_E\rho/\nu}\right] = 60.68\,Mpa.$$

取 $\nu = 1$

$$\sigma_{c,t} = \sigma_{c,0}\left[1 - \frac{\psi\alpha_E\rho/\nu}{1+\alpha_E\rho/\nu}\right] = 2.63\,Mpa$$

(2) 卸载后，设应变变化为 $\Delta\varepsilon$.

则 $N' = (\sigma_{c,t} - E_c\Delta\varepsilon)A_c + (\sigma_{s,t} - E_s\Delta\varepsilon)A_s = 0$. ⇒ 解得 $\Delta\varepsilon = 110.756\mu$

则有 $\sigma_c = \sigma_{c,t} - E_c\Delta\varepsilon = -0.93\,Mpa$ (拉)

$\sigma_s = \sigma_{s,t} - E_s\Delta\varepsilon = 38.53\,Mpa$ (压)

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